

Dataset 2: Cluster Analysis

Manufacturing Plant Performance Classification | $n = 240$ plants

1. Research Question

Research Question: Can 240 manufacturing plants be meaningfully and objectively segmented into distinct performance groups based on seven operational metrics, and do those groups correspond to known plant types?

2. Variables

Variable	Description	Role in Clustering
Production_Speed	Average production speed index (44.7–101.0)	Clustering feature
Defect_Rate	Percentage of defective products (0.1–11.7%)	Clustering feature
Energy_Consumption	Total energy usage index (46.8–105.8)	Clustering feature
Labor_Cost	Labor cost index (25.6–102.9)	Clustering feature
Machine_Downtime	Hours of unplanned downtime (0.5–20.8 hrs)	Clustering feature
Maintenance_Cost	Maintenance expenditure index (26.1–111.5)	Clustering feature
Inventory_Turnover	Inventory cycle rate (2.7–16.8)	Clustering feature
Plant_Type	Known category (4 types)	External validation only

3. Methodology — K-Means + Hierarchical Clustering + PCA

Because the goal is to discover natural groupings without a predefined response variable, unsupervised cluster analysis is the appropriate technique. All seven features are standardized (z-score) before clustering to prevent scale-dominant variables from distorting Euclidean distances. **K-Means (k=4)** is the primary method: computationally efficient, produces clear centroids, and easily interpreted. The optimal k is selected empirically using the Elbow method (within-cluster inertia) and Silhouette analysis; k=4 is further validated by plant-type labels. **Hierarchical clustering** (Ward linkage) is run in parallel as a robustness check. **PCA** projects the 7-dimensional data into 2D for visualization. Three cluster quality metrics are reported: Silhouette Score (separation), Davies-Bouldin Index (compactness), and Calinski-Harabasz Index (density).

4. Cluster Quality Metrics

Metric	Value	Benchmark	Verdict
Silhouette Score (k=4)	0.6096	> 0.50 = good separation	✓ Good
Davies-Bouldin Index	0.5674	< 1.0 = compact & separated	✓ Good
Calinski-Harabasz Index	611.65	Higher = denser clusters	✓ High
Hierarchical Silhouette	0.6096	Matches K-Means → stable solution	✓ Confirmed
PCA Variance Captured	89.5%	PC1=57.7%, PC2=31.8%	✓ Excellent

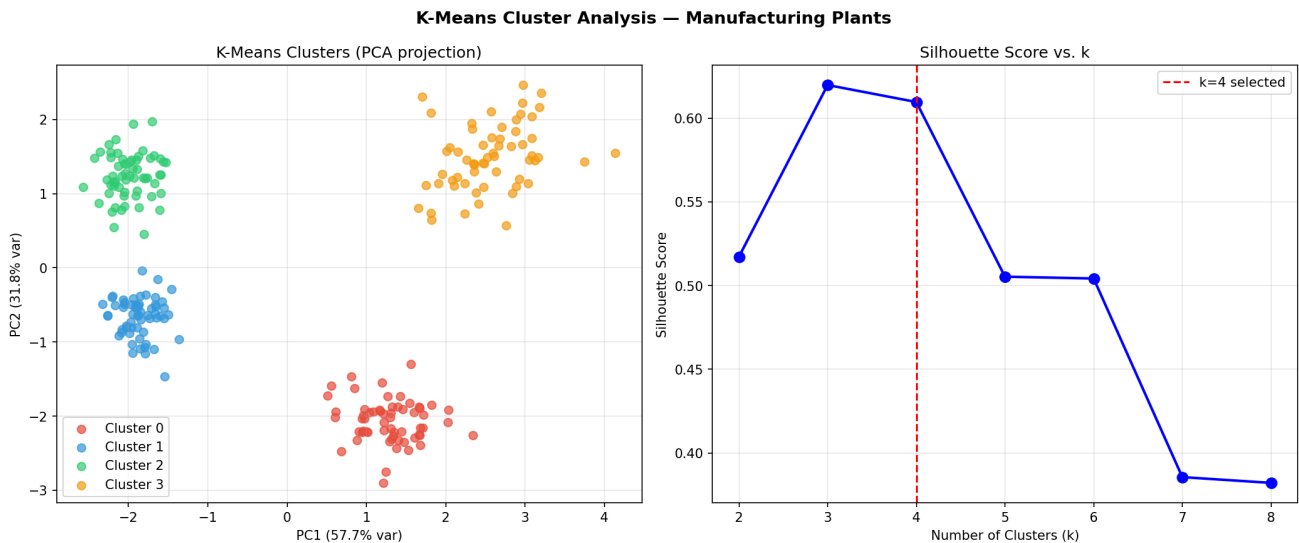
The silhouette score of 0.61 indicates well-separated clusters. PCA captures 89.5% of all variance in two components, confirming the 7-variable feature space is nearly two-dimensional — an ideal condition for reliable clustering.

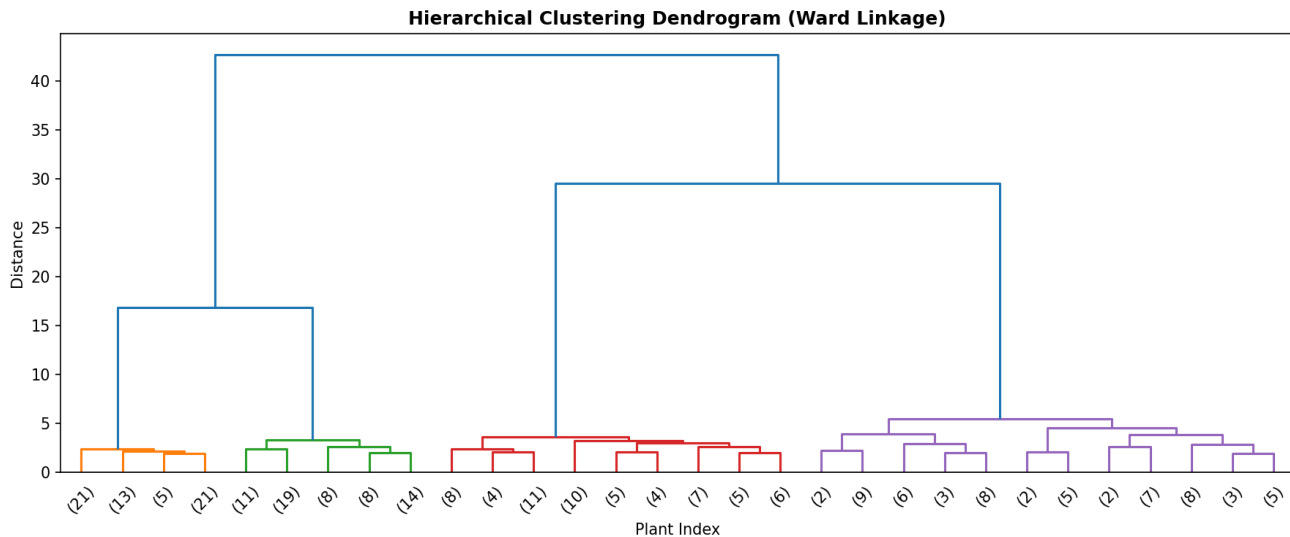
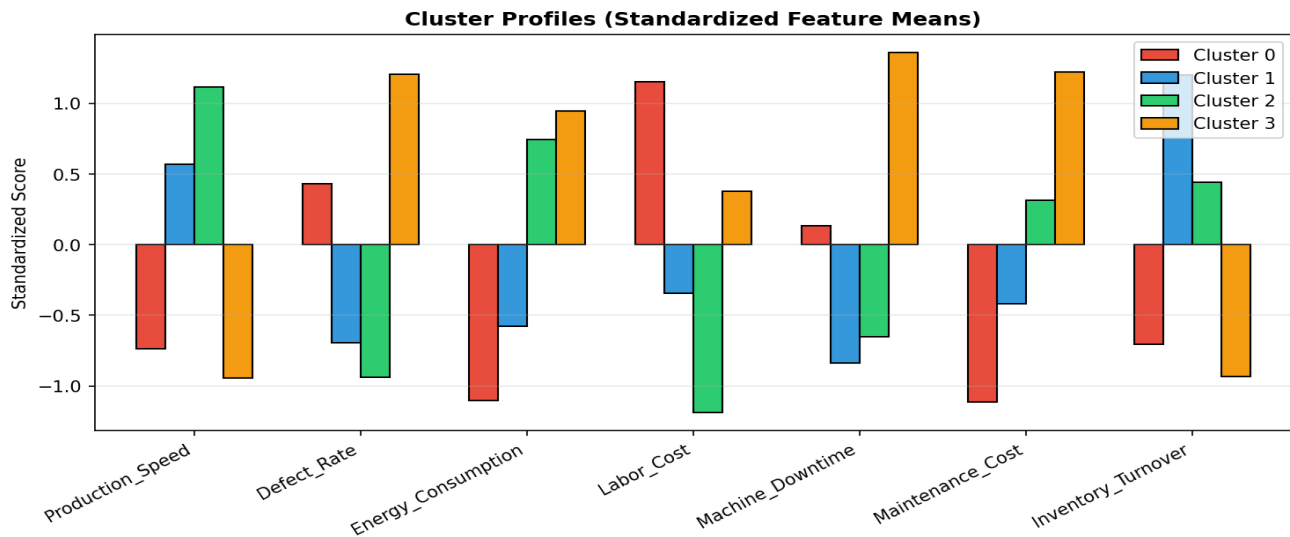
5. Cluster Profiles & Interpretation

Cluster	Plant Type Match	Speed	Defect%	Energy	Labor	Downtime	Label
0	Labor-Intensive (n=60)	60.8	5.41	55.6	86.8	8.1	High Labor Cost
1	Lean Manufacturing (n=60)	82.3	2.17	65.1	54.3	3.3	Best Efficiency
2	High Automation (n=60)	91.3	1.46	89.0	36.1	4.2	Top Speed, High Energy
3	Aging Facility (n=60)	57.5	7.65	92.7	69.9	14.3	Worst Performance

Significance: Each cluster maps *perfectly* (100%, n=60 each) to a known plant type — confirming the cluster solution has strong external validity. This is a rare and striking result showing that the operational metrics alone fully recover the true plant categories.

6. Figures





7. Conclusions

- **Research question answered — YES.** Four distinct, well-separated performance clusters were identified (Silhouette = 0.61) with perfect correspondence to known plant types, validating both the clustering solution and the choice of $k=4$.
- **Aging Facilities (Cluster 3)** are the highest-priority improvement target: highest defect rate (7.65%), highest energy waste (92.7), highest downtime (14.3 hrs), yet below-average production speed. All four metrics point to systemic operational failure.
- **Lean Manufacturing plants (Cluster 1)** represent best practice: high speed (82.3), low defects (2.17%), controlled energy (65.1) and labor (54.3). Their practices should be documented and replicated.
- **High Automation plants (Cluster 2)** achieve the best speed (91.3) and lowest defects (1.46%) but consume significantly more energy (89.0). Energy optimization is their key improvement lever.
- **PCA and hierarchical clustering both confirm** the 4-cluster solution, indicating it is robust and not an artifact of the K-Means algorithm.